

Day 5 — A 5×5 Magic Square

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will independently construct a 5×5 magic square using the *turagagati* method and verify its magic constant of 65.

Materials

- Printed blank 5×5 grids (3 per student)
- The 5-step rule handout from Day 4
- Stopwatch (for timed second-construction)

Vocabulary review

- *turagagati* — stair-step method (introduced Day 4).
- *magic constant* for 5×5 → **65** (from Day 2).

Lesson flow

Warm-up (5 min)

- Class recall: 5-step rule, recited together.
- Magic-constant check: “For a 5×5 with numbers 1–25, the magic constant is ____?” (65.)
- Pre-class question: did everyone produce the *same* 5×5 last night? Most will say yes.

Core (20 min)

1. **Build a 5×5 together on the board.** Narrate every step. Take 8–10 minutes — go cell-by-cell.

Start with 1 in row-1, column-3 (the top middle). Place 2 up-right (off the top → wraps to bottom row, column 4). Place 3 up-right from 2 (off the right → wraps to row 4, column 5)... continue.

Expected result (the classical Nārāyaṇa 5×5):

17		24		1		8		15

23		5		7		14		16

4		6		13		20		22

10		12		19		21		3

11		18		25		2		9

2. **Verify together.** Pick:

- One row $\rightarrow 17 + 24 + 1 + 8 + 15 = 65 \checkmark$
- One column $\rightarrow 17 + 23 + 4 + 10 + 11 = 65 \checkmark$
- The main diagonal $\rightarrow 17 + 5 + 13 + 21 + 9 = 65 \checkmark$
- The anti-diagonal $\rightarrow 15 + 14 + 13 + 12 + 11 = 65 \checkmark$

3. **Notice the structure.** Take 3 minutes to observe:

- 13 (the middle number) is in the centre — consistent with the $n^2/2 + 1 = 13$ rule for odd-order.
- Numbers 1–9 occupy a “broken diagonal” pattern; 16–25 occupy the complementary pattern.
- The same algorithm that gave us the 3×3 gave us this. The rule does not change with size.

Activity (15 min) — Formative quiz Part B + speed construction

- **First 10 min:** Hand out quizzes/formative.md Part B. Students complete individually.
- **Last 5 min:** Each student starts a fresh 5×5 grid. On “Go,” they construct a magic square from scratch as fast as they can. Self-time. Verify one row before stopping the watch.

Wrap-up (5 min)

- Show of hands: who finished in under 4 minutes? Under 6? Under 8?
- “Yesterday’s home practice was the same 5×5 . Tomorrow we look at a different beast — the 4×4 .”
- Preview Day 6: Khajuraho and the 4×4 .

Student-facing content

The 5×5 magic square — classical Nārāyaṇa form

Magic constant: 65 (from $S = n(n^2 + 1)/2$ with $n = 5$).

17		24		1		8		15

23		5		7		14		16

4		6		13		20		22

10		12		19		21		3

11		18		25		2		9

Built using the *turagagati* method, exactly as for the 3×3 — the rule does not change with size, as long as n is odd.

Homework

- Construct a 7×7 magic square using the stair-step method. The magic constant should be 175. Yes, it takes time — work patiently. Bring it tomorrow for spot-checks.

Differentiation

- **One tier up:** Construct a 9×9 magic square. Constant = 369. Time it.
- **One tier down:** Construct another 5×5 , slowly, with the rule sheet visible. Speed is not the goal yet.

Teacher notes

- For the live demonstration, use a slow pace and check student-following frequently — “where would 2 go?” “is that cell filled?”
- Students may resist because “I already built one at home.” Push them to *re-build* it on the board — narrating the rule reinforces it.
- A few students will produce mirror-image squares because they reversed the up-right direction. Show that mirror images are also valid magic squares (by symmetry from Day 3) — but consistency matters when you’re describing the algorithm.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — the 5×5 example is one of the worked examples in the original text.